VIDYAPEETH

BATCH CODE: AN-181MA|2026

SUBJECT NAME: BOTANY

CHAPTER NAME:
Cell – The unit of life

Lecture No. 3

By- Komal Rai

Today's Goal

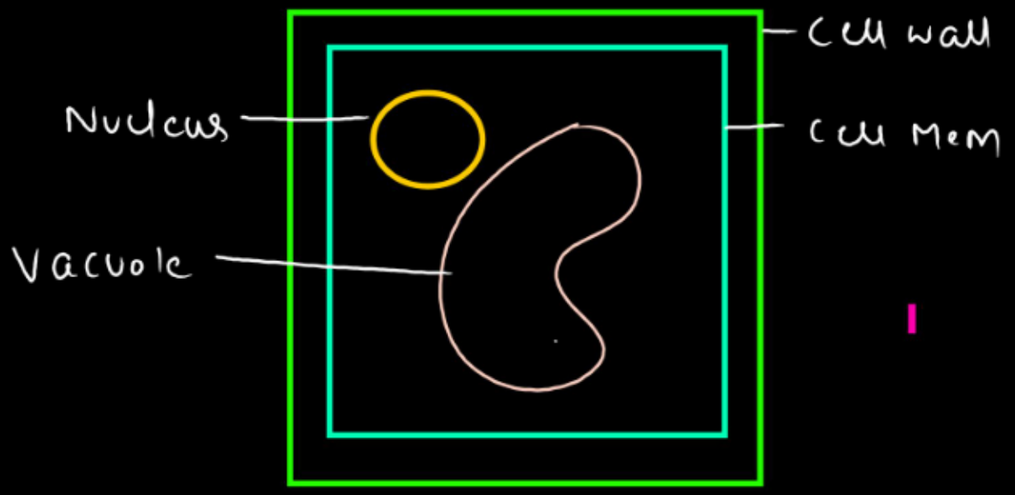
Subtopic .. Overview of cell

Subtopic .. Prokaryotic cell

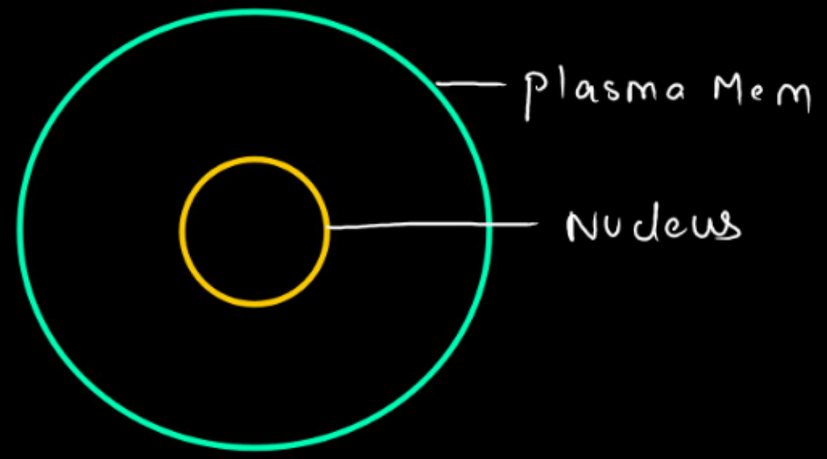
Subtopic ..

Topic: Over view of cell

Plant cell - onion peel



Animal cell - Human cheek cell



Neet

Cytoplasm is a Main arena of all cellular activities (for both plant & Animal cell.

Neet

Human cheek outer Mem is a delimiting Structure of cell (cell Membrane)

→ cells vary in terms of

Size

Shape.

Size

Neet.

Match the column.

1) Smallest living cell — Mycoplasma (0.3 μm in length)

2) Bacteria — 3-5 μm in length — Statement

3) longest cell — Nerve cell

4) largest cell — ostrich Egg

5) Human R.B.C — 7 μm in diameter.

Fig. Match the column

Neet

Typical Eukaryotic cell	10 - 20 μm
Typical Bacteria	1 - 2 μm
P.P.L.O (Pluero Pnuemonia Like organism)	0.1 μm
Virus	0.2 - 0.02 μm

Q) Which of the following is smallest cell.

- Virus
- Mycoplasma
- Bacteria
- P.P.L.O

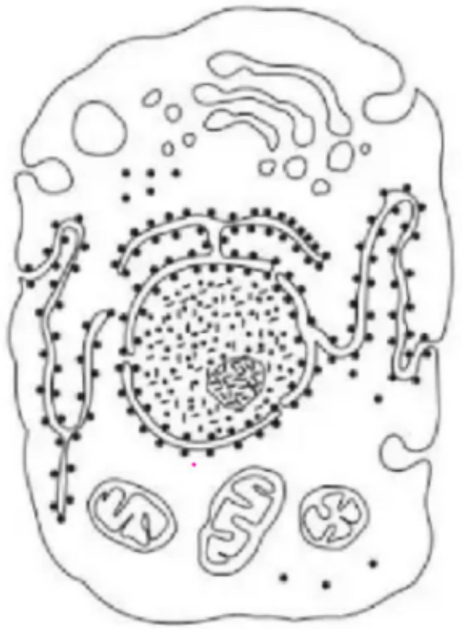
Shape

Neet

Match the column

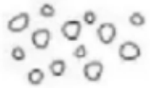
R.B.C	Round / biconcave
W.B.C	Amoeboid
Nerve cell	long / Branched
Columnar Epithelium	long / Narrow
Tracheids	Elongated
Mesophyll cell	Round / oval

Match the column Neet



A typical eukaryotic cell
(10-20 μm)

Typical bacteria
(1-2 μm)



PPLO
(about 0.1 μm)

Viruses
(0.02-0.2 μm)

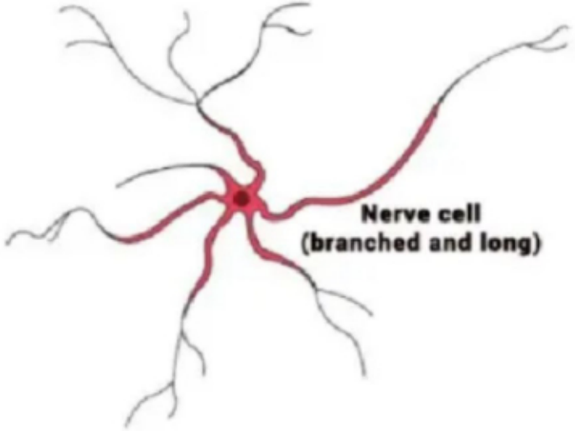
Figure 8.2 Diagram showing comparison of eukaryotic cell with other organisms

Neet

Red blood cells
(round and biconcave)

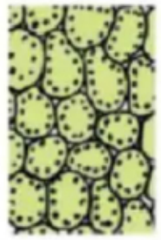
White blood cells
(amoeboid)

Columnar epithelial cells
(long and narrow)



Nerve cell
(branched and long)

A tracheid
(elongated)



Mesophyll cells
(round and oval)

→ Cell is also divided into two types on basis of **Organisation**

1) Prokaryotic cell
2) Eukaryotic cell.

P. cell	E. cell.
Primitive form of cell	True form of cell
Nucleus – Nst N Mem – Nst	Well defined Nucleus i.e N. Mem + Nst
G. Material (DNA) + Nst Cytoplasm Circular	G. Material (DNA) + Nst inside Nucleus linear
Cell organelles – Nst Except Ribosome (70S)	Cell organelle + Nst Ribosome (80S)

Topic: Homework-1

- DPP

DPP NO	Question No
DPP -	
DPP -	

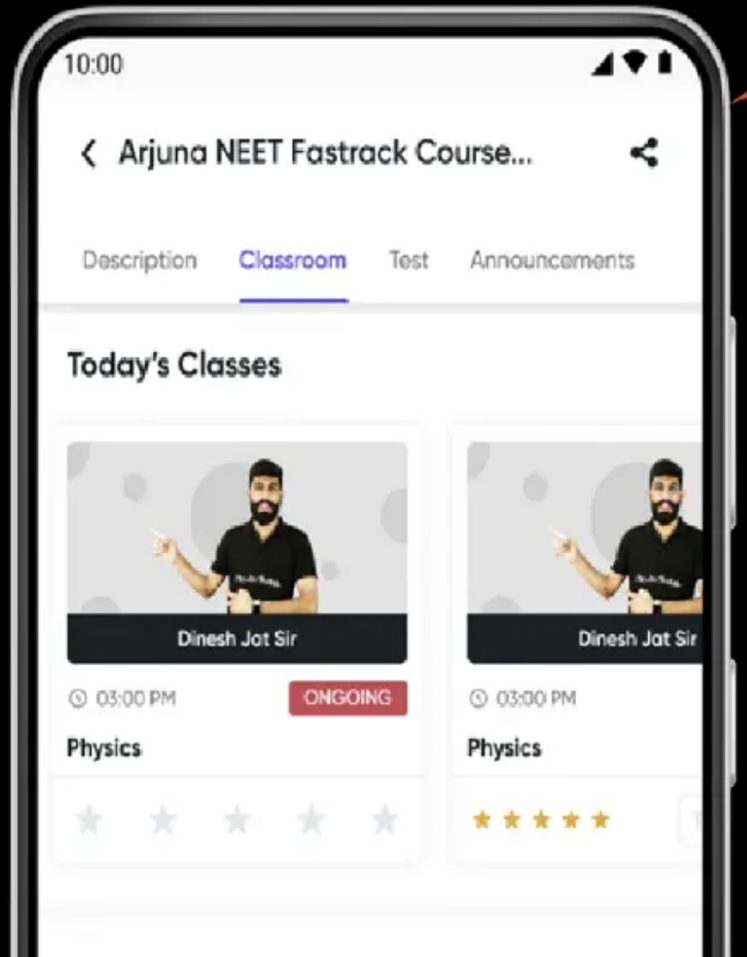
- Module
  - Exercise Name

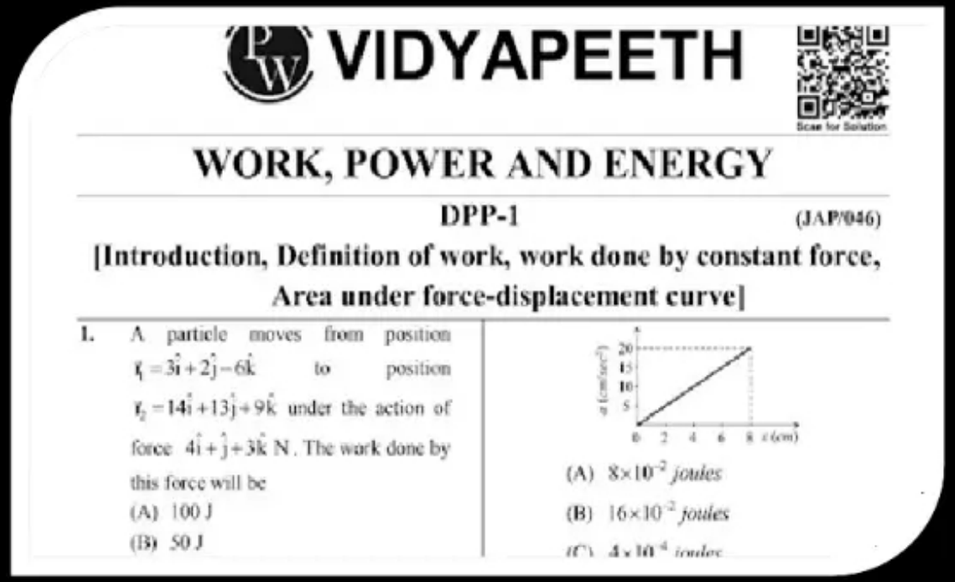
Module Page No	Question No
Page No-	
Page No -	

★★★★★

Rate your today's class in the Batch

Solve the DPP and check Solution





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Scan for Solution

WORK, POWER AND ENERGY

DPP-1 (JAP/046)

[Introduction, Definition of work, work done by constant force, Area under force-displacement curve]

1. A particle moves from position $\vec{r}_1 = 3\hat{i} + 2\hat{j} - 6\hat{k}$ to position $\vec{r}_2 = 14\hat{i} + 13\hat{j} + 9\hat{k}$ under the action of force $4\hat{i} + \hat{j} + 3\hat{k}$ N. The work done by this force will be

(A) 100 J

(B) 50 J

(A) 8×10^{-2} joules

(B) 16×10^{-2} joules

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